

Tung Nguyen

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Professional Experience

Principal Data Engineer, Collins Aerospace (RTX) Jan 2023 - Present

- Led data engineering initiatives for Supply Chain; designed and managed 200+ Airflow, Snowflake, & Postgres pipelines supporting an organization of 600+ business users.
- Served as the team's data architect for SAP ERP-related initiatives; consolidated siloed datasets into a medallion architecture, turning high-effort, ad-hoc analyses into a scalable, analytics-ready data foundation.
- Developed a centralized contracts metadata database, enabling the company to identify \$100M+ per year in assertion opportunities, with \$15M of assertions sent out in the first year of operation.
- Developed an extensible simulation framework for financial analyses; integrated external economic data APIs, internal business logic, and statistical sampling methods to forecast part pricing, eliminating multiple hours of manual work per week, per analyst.
- Built an automated ML pipeline for predicting on-time delivery with daily inference and automated monthly retraining; improved existing model precision from 60% to 71% (+11%), resulting in more accurate supplier risk forecasts.
- Modernized Supply Chain's self-service data platform from an on-prem deployment to a Terraform-managed, distributed architecture using AWS ECS, RDS, DynamoDB, and S3.
- Refactored the team's Airflow deployment from a monolithic deployment to a distributed deployment with independently auto-scaled components using AWS ECS, RDS, and SQS.

Senior Structural Analysis Engineer, Collins Aerospace (RTX) Jun 2018 - Jan 2023

- Designed and analyzed aerospace structures across a wide range of commercial and military platforms.
- Recognized as an Engineer of the Year semi-finalist for establishing structural optimization standards for the company, increasing typical weight margins from 1-5% to 10-20%.
- Built multiple automation tools supporting analysis, optimization, and collaboration; introduced NLP-based methods for solving problems related to unstructured engineering data.

Projects

Agentic Anomaly Detection System

- Built an end-to-end anomaly detection system for multivariate time-series data by combining forecasting algorithms with language models for interpretability.
- Automated context engineering using LangChain tool bindings and a vector database for RAG; deployed system via a lightweight chatbot interface built in Streamlit for interactive anomaly analyses.

Education

Georgia Institute of Technology, MS in Analytics, GPA: 3.9/4.0	2025
University of California - San Diego, MS in Structural Engineering, GPA: 3.9/4.0	2018
University of California - San Diego, BS in Structural Engineering, GPA: 3.8/4.0	2017

Technical Skills

Programming: Python, Docker, Visual Basic, R, C#

Cloud: AWS, Terraform

Data & ETL: Airflow, SQL, Spark, Databricks

Visualization: Tableau, PowerBI, Python